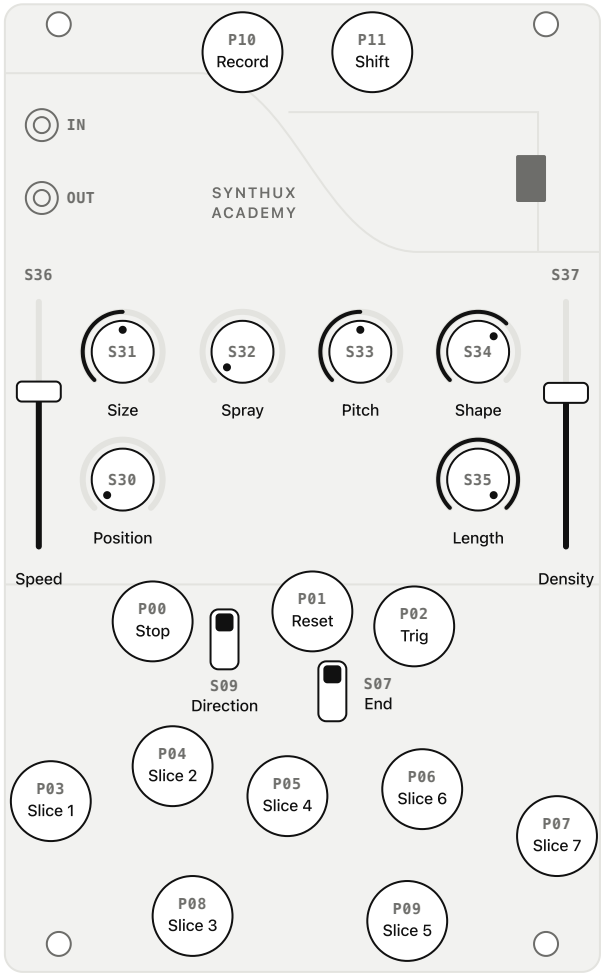
Simple Touch

Clessidra

It's the instrument where you pour grains of sound from single drops into a dense cloud.

francescodimaggio.nl/clessidra

Synthux Design Residency 2026



Record a sound, then play it as grains: Speed through it, Density how they fall.

Two layers: each knob alone, and its twin with P11 held. Nothing jumps when they change.

Density at the bottom: no grains but the slices'. S09 at its centre: frozen.

FADERS AND KNOBS

S36	**Speed**. 0 to 4×, with stops at ¼, ½, 1, 2 and 4×. The pitch stays. Frozen: scrubs the loop. *P11*: **Volume**. Output level, off to 0 dB.
S37	**Density**. Grains per second, 0 to 320. At the top with Size at its centre, all 32 voices sound. *P11*: **Random timing**. When grains start, from a regular pulse to random.
S30	**Position**. Where the loop starts. The playhead moves with it. *P11*: **Filter**. Low-pass below the centre, off at the centre, high-pass above.
S35	**Length**. How much of the recording after Position the loop takes. *P11*: **Reverb**. Reverb on the grains, from none to long tails.
S31	**Size**. Grain length, 5 ms to 2 s, 100 ms at the centre. *P11*: **Random size**. Each grain a different length, up to two octaves either way.
S32	**Spray**. How far grains stray from the playhead, up to the whole loop, and across the stereo field. *P11*: **Reverse**. How many grains play backwards: none, some, all.
S33	**Pitch**. Transpose up to two octaves either way, with stops at musical intervals. *P11*: **Random pitch**. Off at the centre. Above: any pitch up to an octave either way. Below: octaves only.
S34	**Shape**. Grain envelope: square, decay, bell, reverse, and everything between. *P11*: **Random shape**. Each grain a different envelope.

PADS AND SWITCHES

S09	**Direction**. Forward, freeze or reverse. Frozen, Speed scrubs through the loop.
S07	**End**. At the loop's end: play once, turn over and run back, or loop. Slices follow it.
P10	**Record**. Tap: keeps the last 8 s. Hold: records up to 30 s. With Shift: hear the input.
P11	**Shift**. Hold for each knob's second function.
P00	**Stop**. Held: stops as a tape machine, slowing to a halt. Released: back to speed.
P01	**Reset**. Second functions back to their defaults.
P02	**Trig**. Plays the loop from its start, at once.
P03–P09	**Slices**. The loop in seven slices: its start, then the recording's strongest attacks inside it. Without attacks, as in a held note, even parts. Left to right on the board: P03, P04, P08, P05, P09, P06, P07. Held: plays its slice in the direction S09 sets, to its end as S07 sets. Several at once. The loop waits.

TO PLAY

Rhythm	Density low, Random timing (P11 + S37) down, Shape on decay, Size short. Size sets how long each hit lasts
Drops	Density near the bottom, Size short: single grains, now and then
Drums by hand	Density at the bottom, Shape on square: tap the slices, several at once
Cloud	Density at the top, Size 150 to 300 ms, Shape on bell, Spray up, Random timing (P11 + S37) full, Reverb (P11 + S35) up
Stretch	Speed at ¼ or ½, Density up. S09 down to hear it backwards
Drone	S09 at the centre, Density at the top, Size long, a little Spray, Reverb up. Speed scrubs the point
One pour	S07 down: the loop plays once and stops. Trig pours it again
Shimmer	Random pitch (P11 + S33) below the centre: octaves
Backwards syllables	Reverse (P11 + S32) at the top: the phrase in order, each grain backwards

LED

Glow	Grains falling
Flashing	Recording
Pulse	At a stop: unison, speed 1×, 100 ms, filter off, random pitch off
Steady	P11 held: the twin last turned